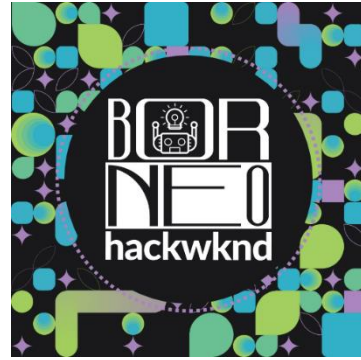




LinguPet

Gamified Indigenous Language Preservation Platform

Aligned with

UN Sustainable Development Goals



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BorNEO HackWknd 2026

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# **1 INTRODUCTION**

## **1.1 Background of Case Study**

Southeast Asia is one of the most linguistically diverse regions in the world. The region contains multiple language families, including Austronesian, Kra-Dai, Austroasiatic, and Sino-Bodic languages. This diversity reflects the complex historical, geographical, and cultural interactions that have shaped the linguistic landscape of the region. While this diversity represents a rich cultural heritage, it also presents significant challenges for communication, education, and the preservation of linguistic traditions.

According to the study *Language Standardization and Its Impact in East and Southeast Asia*, language policies across the region have attempted to address communication barriers through language standardization. For example, Indonesia adopted Bahasa Indonesia as a national language to unify a population that speaks more than 700 local languages. Similarly, several Southeast Asian countries implement bilingual education systems or promote standardized languages to strengthen national identity and improve literacy.

These policies have successfully enhanced national integration and communication efficiency. However, they have also unintentionally contributed to the marginalization of local and indigenous languages. As standardized languages become dominant in education, administration, and digital platforms, younger generations increasingly shift away from traditional languages, which threatens the sustainability of linguistic diversity.

In the era of globalization and digital transformation, indigenous languages face an additional challenge: limited representation in digital spaces. Many languages lack accessible learning platforms, digital documentation, and interactive educational resources. Without technological support, these languages risk gradual decline as intergenerational transmission weakens. Therefore, there is an urgent need for innovative technological solutions that can support both language preservation and language learning. Artificial intelligence and modern digital platforms offer opportunities to create interactive learning systems that help revitalize indigenous languages while making them accessible to younger generations.

## **1.2 Chosen SDG**

This project aligns with Sustainable Development Goal (SDG) 11: Sustainable Cities and Communities, particularly Target 11.4, which aims to protect and safeguard cultural heritage. Indigenous languages are an important part of intangible cultural heritage, carrying the cultural identity, history, and knowledge of local communities.

However, many indigenous languages across the ASEAN region are increasingly threatened by globalization, language shift, and the dominance of national or global languages. Without effective preservation efforts, valuable linguistic and cultural knowledge may gradually disappear.

The proposed platform addresses this issue by creating a digital ecosystem that supports the documentation, preservation, and learning of indigenous languages through community participation and interactive technology. Native speakers can contribute pronunciation recordings and linguistic knowledge, while learners engage with the languages through gamified learning experiences. By encouraging community involvement and intergenerational learning, the platform supports the objective of SDG 11.4 by helping ensure that indigenous languages remain preserved, accessible, and actively used in the digital era.

### **1.3 Problem Statement**

Although Southeast Asia has rich linguistic diversity, many indigenous languages are declining in use, particularly among younger generations. One major factor is the dominance of standardized national and international languages in education and daily communication, which encourages younger people to prioritize widely used languages that offer greater social and economic opportunities. Another challenge is the limited availability of accessible and engaging learning resources for indigenous languages. Most digital language learning platforms focus on globally dominant languages, leaving indigenous languages underrepresented. In addition, many preservation initiatives emphasize documentation, such as dictionaries or archives, rather than interactive learning systems.

Without innovative approaches that integrate language preservation with modern technology, the number of active speakers may continue to decline, potentially leading to the loss of valuable cultural knowledge and heritage.

### **1.4 Objectives**

The objectives of this project are:

- a. Developing an interactive digital platform that allows users to learn native languages in an engaging way.
- b. Integrating artificial intelligence technology to support adaptive learning and language conversation practice.
- c. Enabling native speakers to contribute linguistic knowledge such as vocabulary, pronunciation, and cultural context to collaborative language datasets.
- d. Encourage young people to learn and interact with native languages through gamification and interactive learning mechanisms.

## **2 REVIEW OF EXISTING SOLUTIONS**

Several digital platforms have been developed to support language learning and cultural preservation. However, most existing solutions primarily focus on widely spoken global languages and do not adequately address the preservation and learning of indigenous languages.

One widely known platform is Duolingo, which offers a gamified learning experience through interactive exercises and reward systems. While the platform successfully increases user

engagement, it mainly focuses on globally dominant languages such as English, Spanish, and Mandarin, providing very limited support for indigenous or minority languages.

Another platform, Memrise, emphasizes vocabulary learning using spaced repetition techniques and user-generated content. Although it allows some community contributions, the learning approach primarily focuses on vocabulary memorization and provides limited cultural context and interaction.

Recent technological efforts have also explored AI-based approaches for language preservation. One example is NightOwlGPT, which aims to support low-resource and endangered languages through artificial intelligence. While this approach shows potential for language documentation and translation, such systems are still relatively new and remain limited in accessibility for broader communities.

In addition to learning platforms and AI initiatives, several projects focus on documenting endangered languages through digital archives and linguistic databases. These repositories store vocabulary, audio recordings, and grammatical descriptions. However, they are often designed primarily for researchers rather than general users, and therefore lack interactive learning features.

From this analysis, several key gaps can be identified. First, there is a lack of gamified learning platforms specifically designed for indigenous language education. Second, many systems do not allow native speakers and local communities to actively contribute linguistic knowledge. Third, cultural representation is often limited, with language learning reduced to vocabulary memorization rather than meaningful communication. These limitations highlight the need for a platform that integrates language preservation, community participation, and engaging learning experiences. By leveraging artificial intelligence and interactive learning technologies, the proposed system aims to create a more inclusive environment where indigenous languages can be preserved and learned by broader communities, particularly younger generations.

### 3 METHODOLOGY

#### 3.1 Tech Stack

The system is developed using this following technologies:

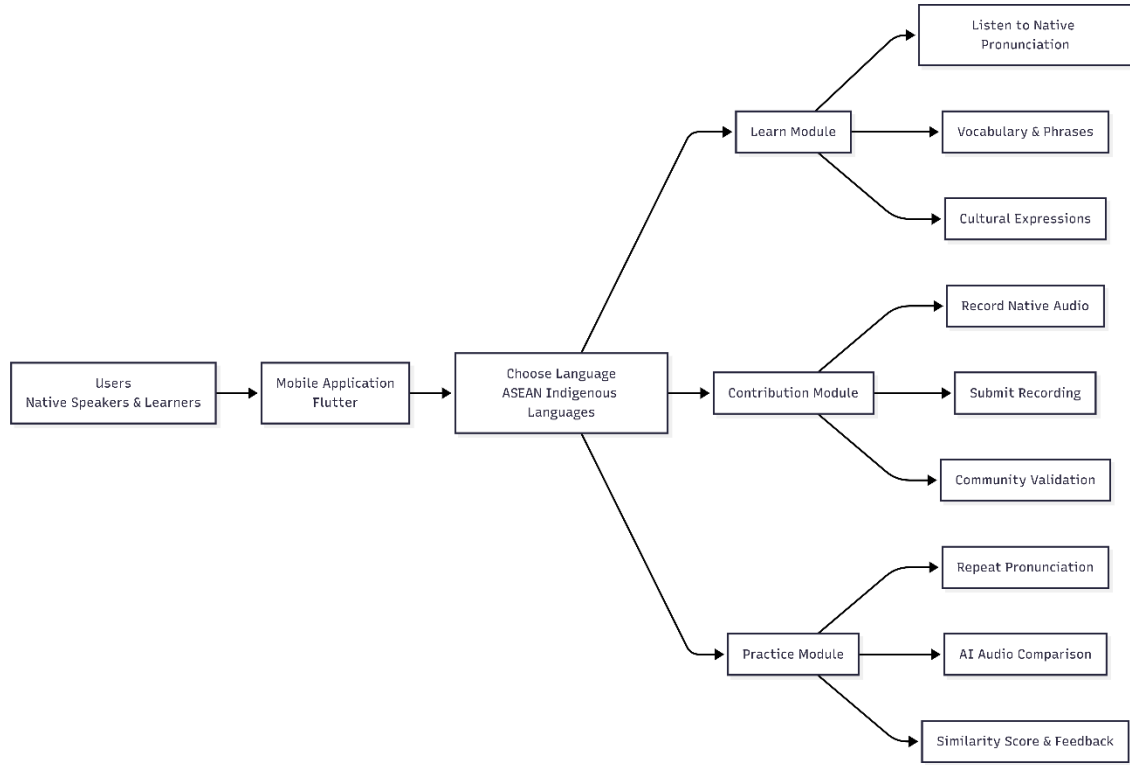
| Component        | Technology                                   |
|------------------|----------------------------------------------|
| Frontend         | Flutter                                      |
| Backend API      | Express.js (TypeScript)                      |
| AI Service       | FastAPI (Python)                             |
| Database         | PostgreSQL                                   |
| ORM              | Prisma                                       |
| File Storage     | Cloud-based storage (e.g., Cloudinary)       |
| Audio Processing | Python libraries such as Librosa and FastDTW |

### 3.2 System Architecture

The system adopts a lightweight microservices architecture consisting of several interconnected components. The frontend is implemented as a mobile application developed using Flutter. This component serves as the primary user interface where both native speakers and learners interact with the platform.

The backend is built using Express.js with TypeScript, functioning as the main API server that handles user requests, manages language datasets, and coordinates interactions between system components. Data storage is managed using PostgreSQL, which stores structured data such as vocabulary, user progress, language levels, and contribution status. Prisma ORM is used to simplify database management and ensure efficient communication between the backend and the database.

Audio files contributed by users are stored using a cloud-based file storage service such as Cloudinary. This approach is used because relational databases such as PostgreSQL are not designed to store raw audio files efficiently. The artificial intelligence component operates as a separate microservice developed using FastAPI in Python. This AI service is responsible for processing audio data and evaluating pronunciation similarity between native speaker recordings and learner submissions. Communication between system components is handled through RESTful APIs, enabling scalable and modular integration between the frontend, backend, and AI services.



*Figure 3.1 System Architecture of the LinguPet Platform*

Figure 3.1 illustrates the functional architecture of the LinguPet mobile application. The diagram shows how users interact with the platform through the mobile application, beginning with language selection and continuing into three main modules: learning pronunciation through native audio examples, contributing recordings to enrich the language dataset, and practicing pronunciation with AI-based evaluation and feedback.

### 3.3 Implementation Process

The implementation of the proposed system follows a structured workflow that ensures both the quality of linguistic data and the effectiveness of the pronunciation learning process.

First, the system establishes a baseline language dataset that serves as a training reference. This dataset contains predefined linguistic elements such as vocabulary, sentences, greetings, and commonly used expressions for each indigenous language. These entries act as prompts that guide contributors in providing pronunciation recordings.

Second, native speakers participate as contributors by submitting audio recordings corresponding to the predefined dataset entries. For example, if the dataset includes the phrase “welcome,” a contributor records its pronunciation in a specific indigenous language, such as Tagalog or other ASEAN local languages.

Third, a community-based review process is conducted to ensure the quality and authenticity of the recordings. Reviewers who are familiar with the language evaluate whether the pronunciation

accurately reflects the correct accent, intonation, and linguistic characteristics of the native language.

Once approved by the reviewers, the validated audio recordings are stored in a cloud-based audio library. These recordings become part of the official language dataset and serve as reference material for the pronunciation learning system. During the learning process, the stored audio recordings function as pronunciation benchmarks. When learners submit their own pronunciation attempts, the system compares the learner's voice input with the reference recordings by analyzing acoustic features such as pitch, intonation, and stress patterns.

Additionally, the system implements a gap analysis mechanism to identify missing audio entries within the language dataset. The AI system automatically detects dataset items that do not yet have pronunciation recordings and prioritizes them as contribution tasks for native speakers. This approach ensures that contributors focus on completing missing linguistic data and gradually enrich the language dataset.

Through this workflow, the system creates a collaborative ecosystem where native speakers contribute authentic linguistic knowledge while learners benefit from accurate pronunciation references. The system includes several core features designed to support indigenous language preservation and interactive learning:

a. Native Speaker Contribution System

Native speakers can contribute linguistic data such as vocabulary entries and pronunciation recordings. These contributions help expand the language dataset and support the preservation of indigenous languages.

b. Language Pet Evolution Mechanism

Each indigenous language is represented by a virtual pet. As more validated linguistic contributions are added to the dataset, the corresponding pet evolves into higher stages, visually representing the growth of the language.

c. Community Validation System

To ensure data quality, user contributions undergo a community validation process. Other users review and approve submitted recordings before they are officially integrated into the language dataset.

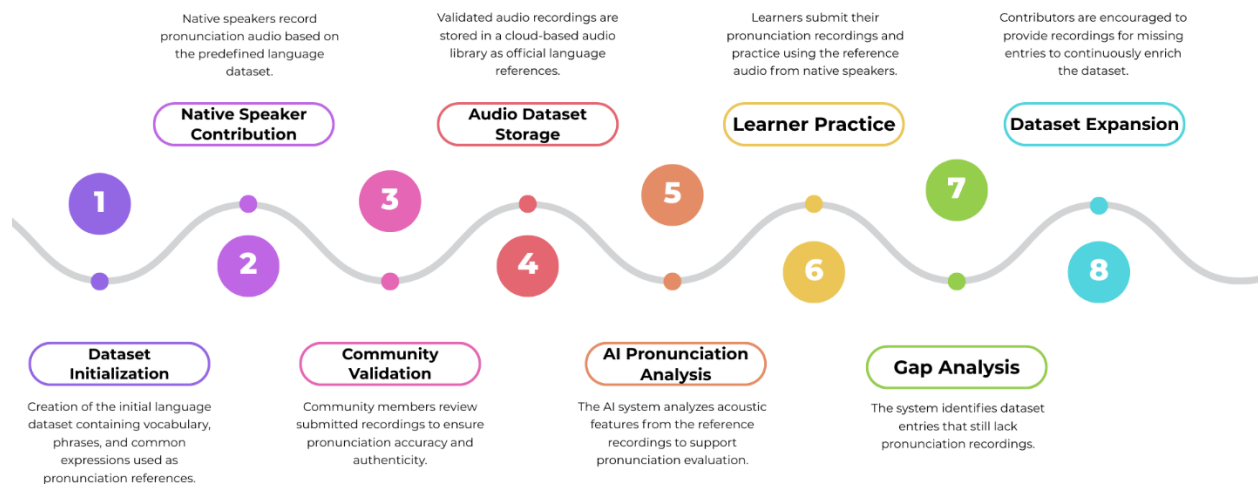
d. Gamified Learning and Taming System

Learners interact with language pets through gamified learning activities. By completing lessons and practicing pronunciation, users gradually tame the corresponding language pet, reflecting their learning progress.

e. AI-assisted Pronunciation Evaluation



The platform integrates AI-based speech processing to compare learner pronunciation with native speaker audio. The system analyzes acoustic features and provides a similarity score that helps learners improve their pronunciation.



*Figure 3.2 Implementation Process of the LinguPet System*

Figure 3.2 illustrates the implementation workflow of the LinguPet system, highlighting the interaction between dataset creation, native speaker contributions, community validation, AI pronunciation analysis, and learner interaction.

## 4 RESULTS

The development of the proposed system resulted in a functional prototype that demonstrates the core concept of LinguPet as a platform for indigenous language preservation and learning. The prototype successfully integrates the main components of the system, including the contributor workflow for native speakers, the validation mechanism for linguistic data, the gamified learning experience for users, and the AI-assisted pronunciation comparison process. Through the implemented prototype, the system is able to illustrate how indigenous language datasets can be gradually expanded through community contributions. Native speakers can submit pronunciation recordings based on predefined linguistic entries, while reviewers ensure that the recordings accurately reflect the authentic pronunciation and linguistic characteristics of the language. Once validated, these recordings are stored as reference materials within the platform's audio library.

The prototype also demonstrates the interactive learning mechanism for users who wish to learn indigenous languages. Learners are able to listen to native speaker recordings and submit their own pronunciation attempts, which are then compared with reference audio to evaluate similarity in terms of pitch, intonation, and stress patterns. This process illustrates the feasibility of using audio-based analysis to support pronunciation learning. In addition, the system successfully implements a gamified interaction model where each indigenous language is represented as a virtual pet that evolves as the language dataset grows. This mechanism visually represents the collective progress of language preservation while simultaneously motivating both contributors and learners to participate in the platform.

Overall, the prototype demonstrates that the proposed approach is capable of combining language documentation, community collaboration, and interactive learning within a single digital ecosystem. These results indicate that the system is able to meet the primary objectives of encouraging indigenous language contribution, supporting pronunciation-based learning, and promoting cultural preservation through engaging digital interaction.

## 5 PROTOTYPE SHOWCASE

### 5.1 Home Page

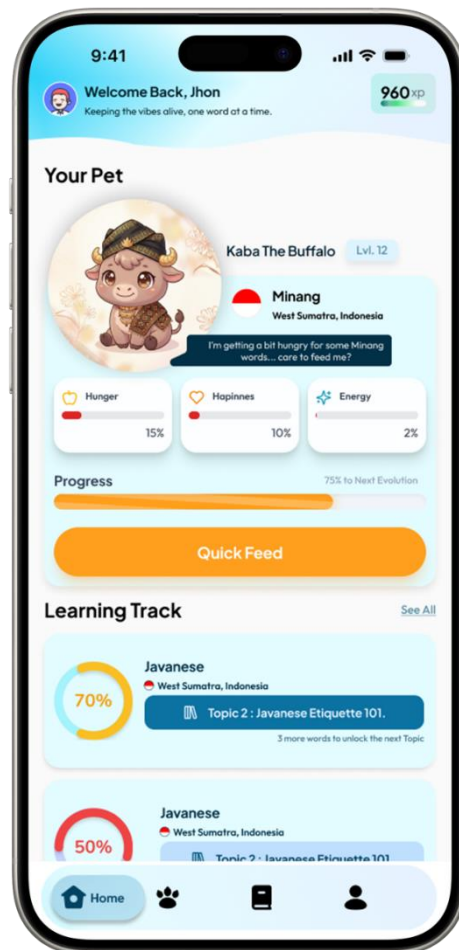


Figure 5.1 Home Page Interface with Pet Widget and Learning Progress Overview

The Home Page serves as the main dashboard of the application. It contains the Pet Widget and the Learning Track, which represent the core interaction between learning and pet-taming.

The Pet Widget displays the user's virtual pet along with its basic information such as the pet's name, level, and origin. It also shows the pet's status indicators, including hunger, happiness, and energy, which reflect the pet's condition. Users can maintain the pet by using the Quick Feed feature.

Below the pet section, the Learning Track shows the user's progress in learning while taming the pet. It displays the current language track, the topic being studied, and the percentage of progress. This section helps users monitor their learning development and track how close they are to unlocking the next topic or stage.

## 5.2 Virtual Pet Interface

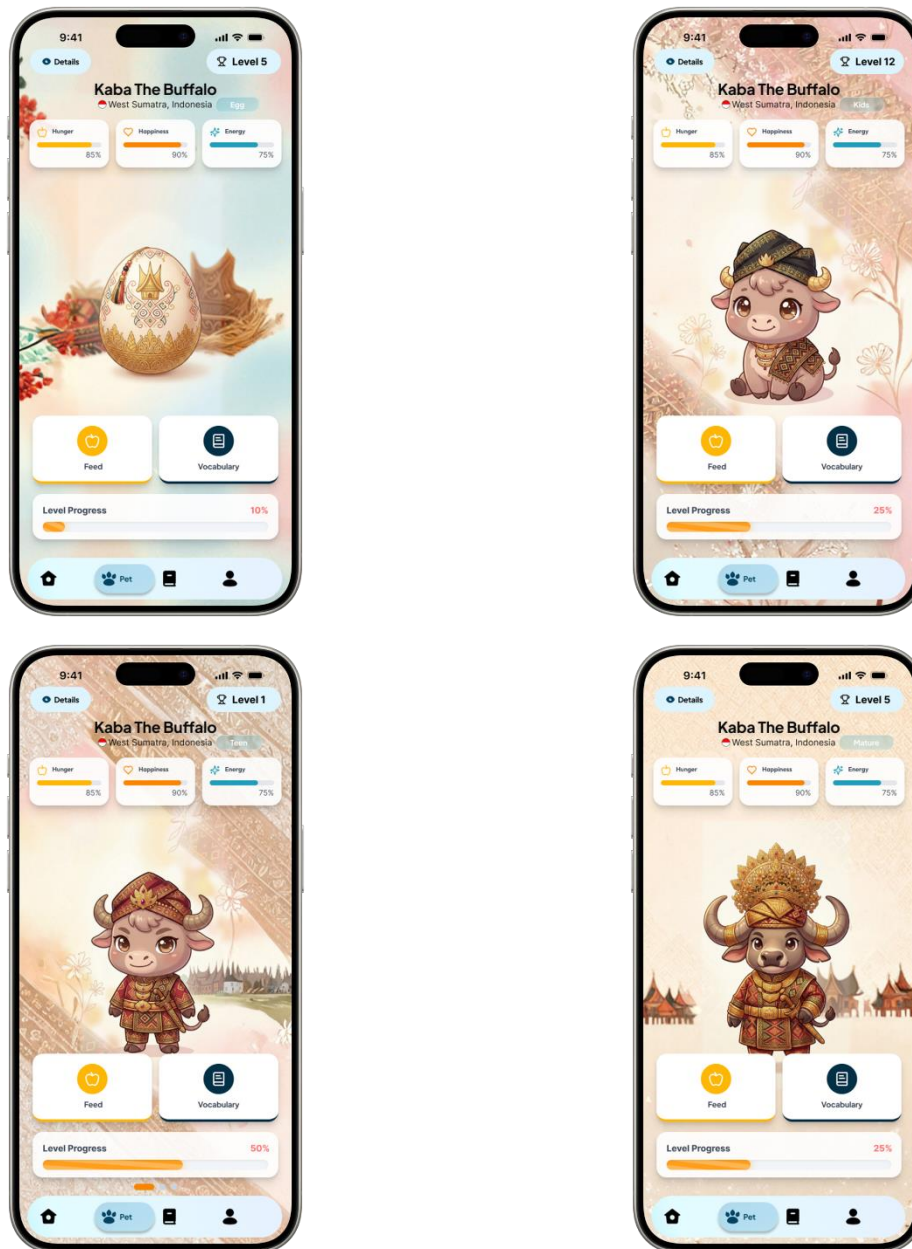
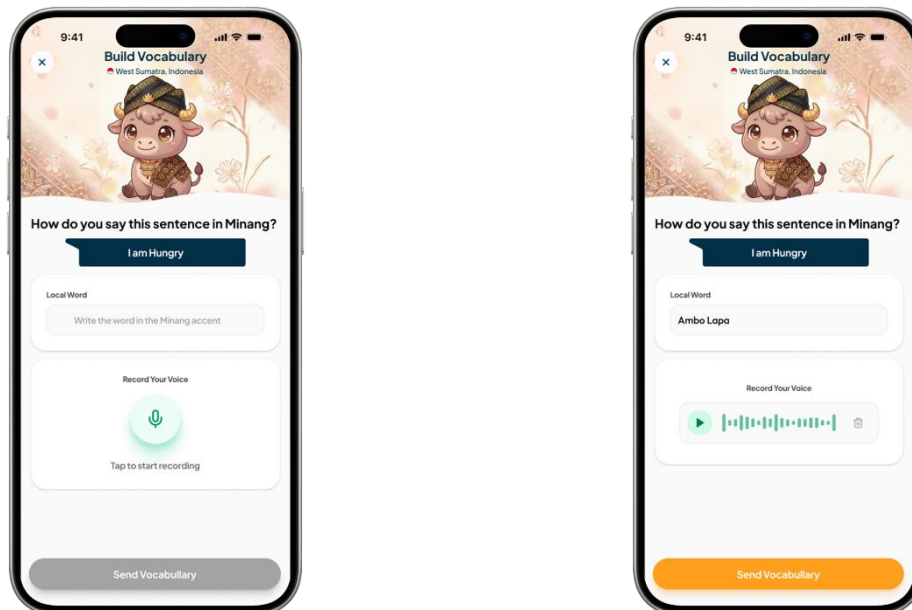


Figure 5.2 Virtual Pet Interface Showing Pet Status, Level, and Language Progress

The Virtual Pet Interface displays key information about the user's virtual pet, including the pet's name, origin, and level. It also shows the overall level progress, which indicates how close the pet is to reaching the next level in its development.

In addition, the interface presents the pet's mood statistics, such as hunger, happiness, and energy, which reflect the pet's current condition. Users can interact with the pet through actions like feeding and accessing the pet's vocabulary, while the interface also shows the pet's current status and visual appearance as it evolves over time.

### 5.3 Vocabulary Building Page

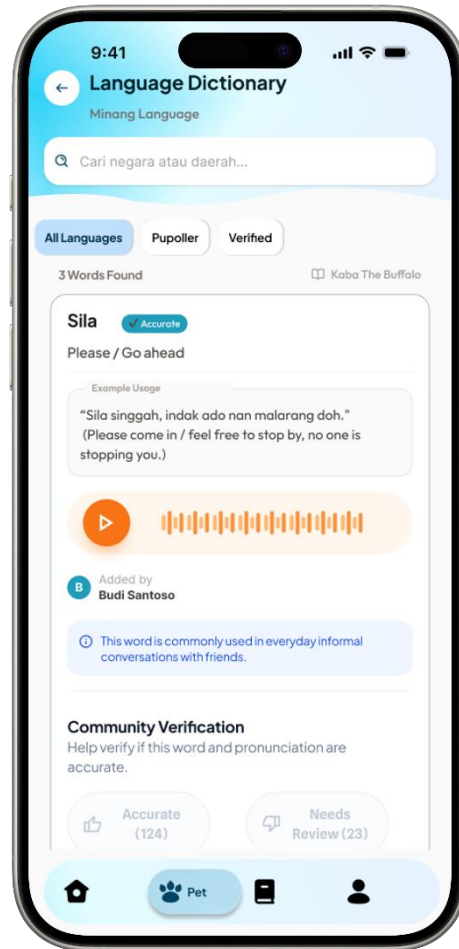


*Figure 5.3 Vocabulary Building Page for Adding New Words and Pronunciation Data*

The Vocabulary Building Page serves as the main interaction space where users actively contribute to preserving regional languages. On this page, users are asked to translate a sentence into the selected local language and input the text into the provided field.

In addition, users can record their voice pronunciation of the translated sentence. Both the text input and voice recording are then submitted as vocabulary contributions, allowing users to participate directly in documenting and supporting the preservation of local languages.

## 5.4 Language Dictionary Page



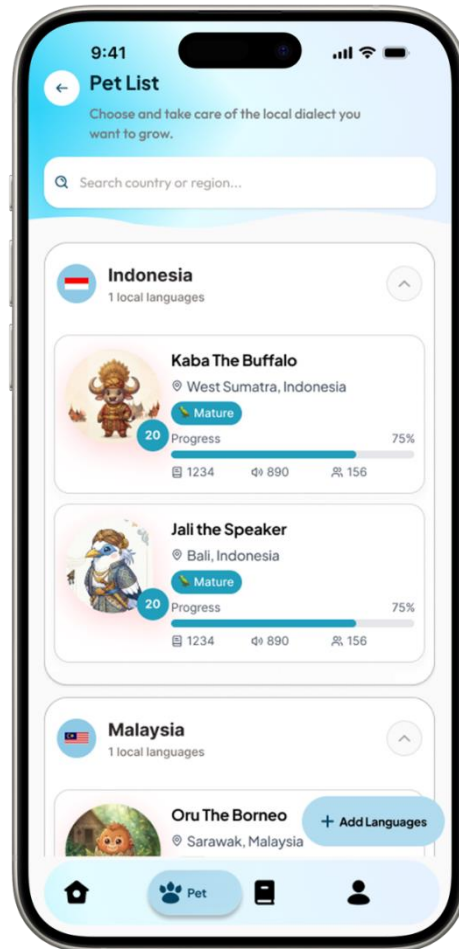
*Figure 5.4 Digital Language Dictionary Displaying Collected Indigenous Vocabulary*

The Language Dictionary page functions as a digital library that displays a collection of indigenous language vocabulary gathered from native speaker contributions. This feature allows users to browse various words and expressions that represent the linguistic resources available in the platform.

Each vocabulary entry includes the written form of the word along with its corresponding pronunciation recording contributed by native speakers. This page serves as a central reference for learners to explore indigenous language vocabulary and listen to authentic pronunciation examples.

By organizing contributed linguistic data into an accessible digital dictionary, the platform supports both language learning and the preservation of regional linguistic knowledge.

## 5.5 Pet Selection Page

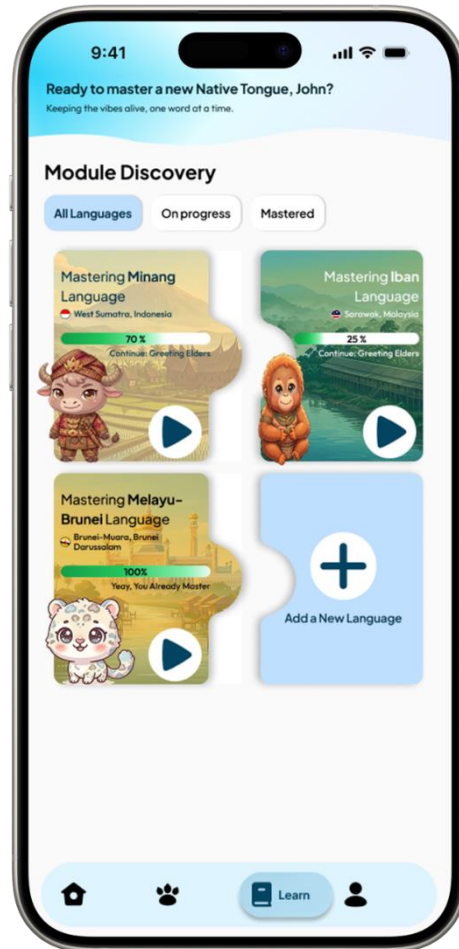


*Figure 5.5 Pet Selection Interface for Choosing Indigenous Language Guardians*

The Pet Selection Page displays a list of available pets associated with different countries or regional languages. Each pet represents a specific culture or language that users can choose as part of their learning journey.

This page also includes an “Add Languages” button that allows users to add new languages, which will unlock additional pets related to those regions. This feature enables users to expand their learning options and interact with more virtual pets from different cultures.

## 5.6 Module Discovery Page

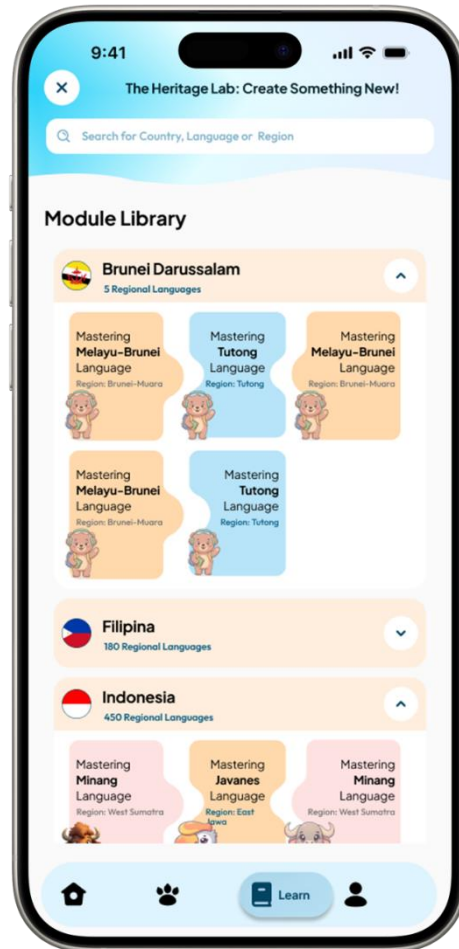


*Figure 5.6 Module Discovery Interface Showing Active Language Learning Modules*

The Module Discovery page serves as the central learning hub where users can explore and manage the indigenous language modules they are currently studying. Each module is represented by a language pet, allowing users to easily monitor their learning progress through a visual and gamified interface. This page also provides an option for users to add new indigenous language modules they wish to learn, enabling continuous exploration of different regional languages within the platform.



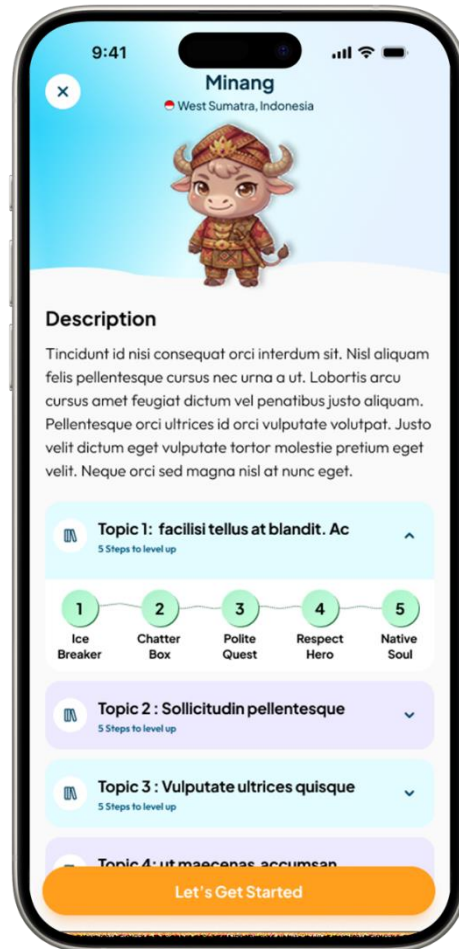
## 5.7 Explore Learning Page



*Figure 5.7 Explore Learning Catalog for Browsing Language Modules by Country*

The Explore Learning page functions as a central catalog where users can explore various indigenous language learning modules available in the platform. The modules are organized based on their country of origin, allowing users to easily browse and select languages they are interested in learning. This feature helps users discover new regional languages and expand their learning experience within the LinguPet platform.

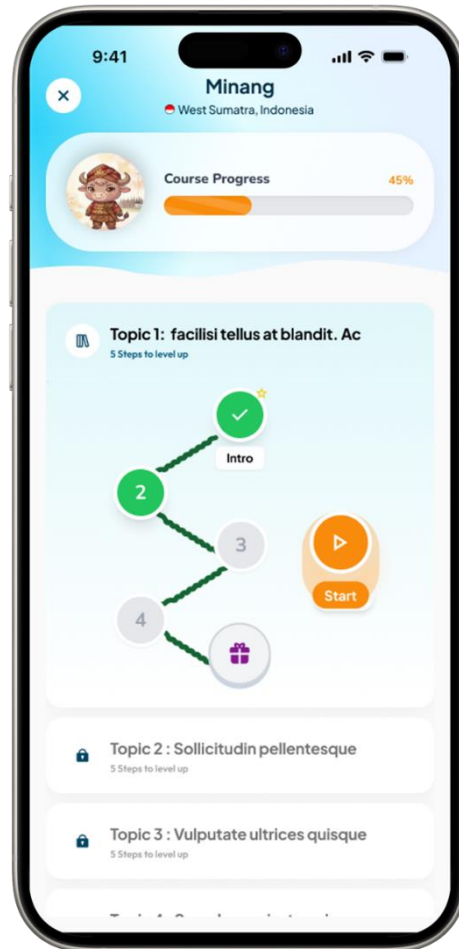
## 5.8 Module Detail Page



*Figure 5.8 Module Detail Page showing topic overview, lesson list, and interactive activity preview.*

The Module Detail page provides comprehensive information about a selected indigenous language learning module. This page presents an overview of the module through a brief description and a structured learning roadmap that outlines the topics covered in the module. Through this interface, users can understand the learning objectives and progression before starting the module.

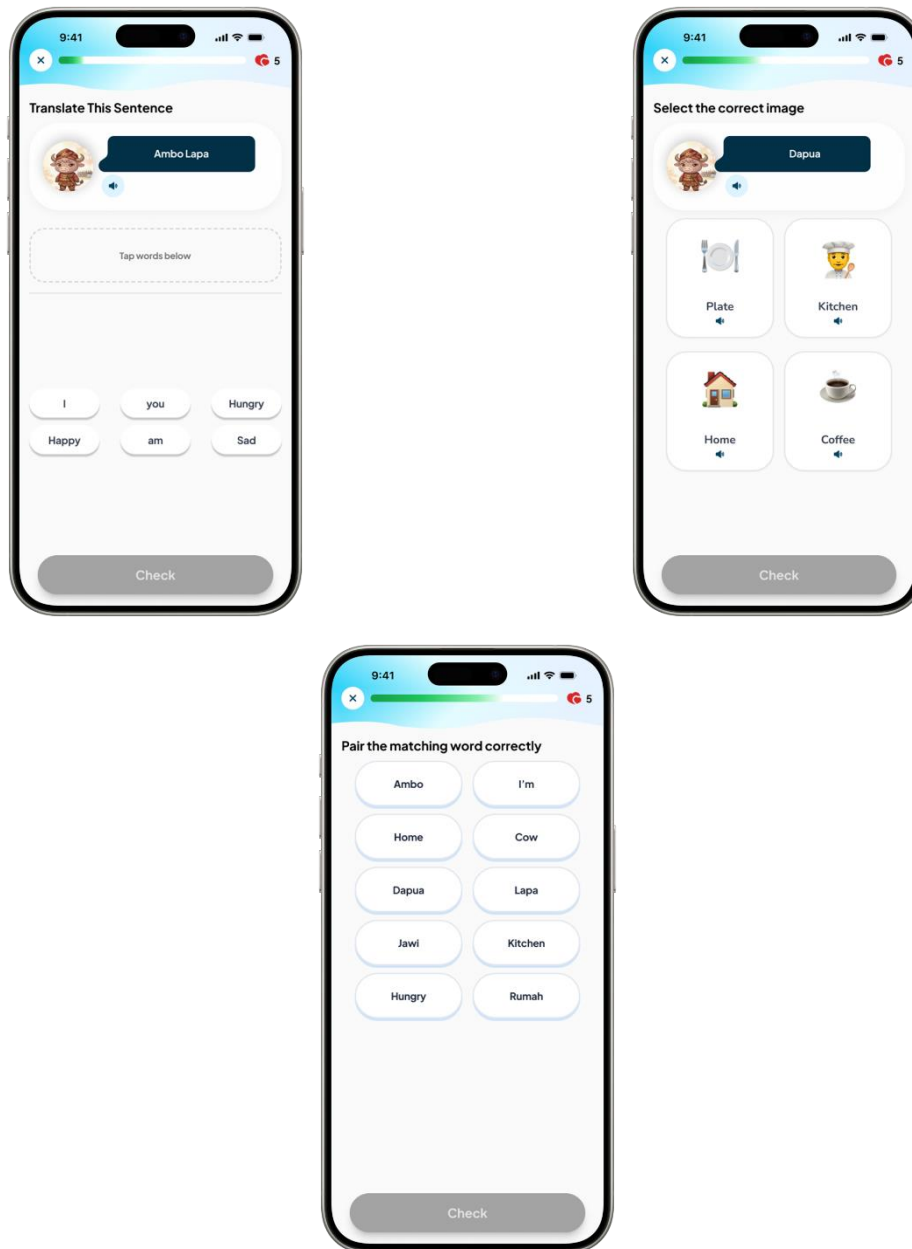
## 5.9 Topic Progress Path



*Figure 5.9 Topic Progress Path Showing Sequential Learning Stages within a Module*

The Topic Progress Path page presents an interactive visualization of the learning journey within a specific topic. This interface guides users through a sequence of learning materials that must be completed in order. By following this structured path, users can progress step by step from one lesson to the next while clearly monitoring their advancement within the topic.

## 5.10 Learning Exercise



*Figure 5.10 Interactive Learning Exercises Including Sentence Builder and Matching Activities*

The Learning Exercise page presents interactive activities designed to help users practice and reinforce their understanding of indigenous language vocabulary and sentence structures. This page includes several types of exercises that encourage active learning. The Sentence Builder exercise requires users to translate a sentence spoken by the pet by arranging randomly provided word fragments into the correct order. The Picture–Word Association exercise asks users to select the image that best represents the displayed word. Meanwhile, the Matching Pairs exercise requires users to match words in the indigenous language with their corresponding meanings in the language selected by the user. Through these activities, the platform provides a gamified learning

experience that helps users improve their comprehension and retention of indigenous language vocabulary.

## 6 AI ACKNOWLEDGMENT

This project utilized several artificial intelligence tools as supportive instruments during the ideation, design, and development stages. In accordance with the competition guidelines, all AI tools used in this project are disclosed transparently to ensure ethical and responsible use of AI technologies.

**ChatGPT** was used primarily to assist in early-stage brainstorming and conceptual refinement. The tool supported the team in exploring possible approaches for indigenous language preservation, structuring initial documentation, and clarifying technical ideas during the proposal development process. Additionally, it was occasionally used to assist in reviewing explanations related to system architecture and development planning.

**Gemini** was utilized to support several development-related tasks, including the generation of visual assets and assisting with the drafting of preliminary code structures. In particular, it was used to help prototype certain interface elements and support debugging during the development of interactive features such as the virtual pet evolution mechanism.

It is important to note that all AI tools mentioned above were used strictly as auxiliary tools to assist the development process. The core concept, system architecture, technical design, dataset workflow, and overall implementation strategy of the proposed platform were independently designed and developed by the project team.

## 7 CONCLUSION

This project introduces LinguPet, a gamified platform designed to support the preservation and learning of indigenous languages across the ASEAN region. By integrating community participation, AI-assisted pronunciation analysis, and a virtual pet system, the platform creates a collaborative environment where native speakers contribute linguistic knowledge while learners practice and improve their language skills. Through the use of mobile technology, cloud-based infrastructure, and audio analysis, LinguPet demonstrates how digital systems can support the preservation of linguistic heritage while providing an engaging and accessible learning experience.

Future development of the platform focuses on improving scalability and expanding the system's ecosystem. The backend architecture can evolve toward an asynchronous microservices model using message brokers such as RabbitMQ or Kafka to efficiently handle large volumes of audio submissions. As the crowdsourced dataset grows, the system can also transition toward more advanced deep learning models such as LSTM networks or fine-tuned speech models like Wav2Vec 2.0 to enable more accurate pronunciation analysis and potential speech-to-text transcription for indigenous languages. In addition, LinguPet can expand through community-

driven curriculum development, enhanced gamification features such as contributor leaderboards and advanced pet evolution stages, and potential Data as a Service (DaaS) opportunities by providing licensed access to collected indigenous speech datasets for research and technological development. Through these improvements, LinguPet has the potential to evolve into a scalable digital ecosystem that contributes to both cultural preservation and the advancement of language technology for underrepresented languages.

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